

ing voltage and current values will be displayed in the voltage and current edit boxes. Note that the program does not take reverse leakage current and dynamic resistance into account in the characteristic curve

9. Click 'close' button to close 'char-

acteristics' window

10. Click 'help' button to open 'help' window

11. Repeat the above steps for different zener numbers selected from the combo box

12. Click 'exit' button to exit the

program.

EFY note. The source code files of this article have been included in this month's EFY-CD.

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FORM1.FRM

'Form Form1.frm is the startup form. OLE 'is used to draw circuit diagram. Global 'variables 'are declared as below

'Declaration of global Variables

```
Dim COMBO1 As Integer
Dim IO As Double
Dim Iz As Double
Dim Vz As Double
Dim IZMAX As Double
```

'When form is loaded, width, height and 'position of form is decided and zener 'numbers are 'added in combobox.

Private Sub Form_Load()

```
'Setting height and width of a form
frm_setupzp.Height = Screen.Height
frm_setupzp.Width = Screen.Width
frm_setupzp.Top = 0
frm_setupzp.Left = 0
```

' adding zener numbers in combobox

```
COMBO1.AddItem "1N4728A"
COMBO1.AddItem "1N4730A"
COMBO1.AddItem "1N4731A"
COMBO1.AddItem "1N4732A"
COMBO1.AddItem "1N4733A"
End Sub
```

'To vary input supply Vscroll1 is used. 'Range of Vscroll1 is adjusted from 0 to '20 in steps 'of 0.1.

'Calculations for finding current through 'zener and voltage across zener are done 'using 'calculation () function.

```
Private Sub VScroll1_Change()
'Adjusting input Voltage range
```

```
VScroll1.Max = 0
VScroll1.Min = 200
Text_Input.Text = VScroll1.Value / 10
If Text_Zresistance.Text = "" Then
Beep
MsgBox ("ENTER VALUE OF RESISTANCE", 32)
Else
If COMBO1.Text = "SELECT" Then
Beep
MsgBox ("SELECT ZENER DIODE"), 32
Else
```

'Calling calculation function according 'to selection of zener

```
COMBO1 = COMBO1.ListIndex
If COMBO1 = "0" Then
zener = Array(100 * 10 ^ (-6), 3.3, 76 * 10 ^ (-3))
IO = zener(0)
Vz = zener(1)
```

```
IZMAX = zener(2)
e = calculations()
Else
If COMBO1 = "1" Then
zener = Array(100 * 10 ^ (-6), 3.6, 69 * 10 ^ (-3))
IO = zener(0)
Vz = zener(1)
IZMAX = zener(2)
e = calculations
```

```
Else
If COMBO1 = "2" Then
zener = Array(50 * 10 ^ (-6), 3.9, 64 * 10 ^ (-3))
IO = zener(0)
Vz = zener(1)
IZMAX = zener(2)
e = calculations
```

```
Else
If COMBO1 = "3" Then
zener = Array(10 * 10 ^ (-6), 4.3, 58 * 10 ^ (-3))
IO = zener(0)
Vz = zener(1)
IZMAX = zener(2)
e = calculations
```

```
Else
If COMBO1 = "4" Then
zener = Array(10 * 10 ^ (-6), 4.7, 53 * 10 ^ (-3))
IO = zener(0)
Vz = zener(1)
IZMAX = zener(2)
e = calculations
```

```
Else
If COMBO1 = "5" Then
zener = Array(10 * 10 ^ (-6), 5.1, 49 * 10 ^ (-3))
IO = zener(0)
Vz = zener(1)
IZMAX = zener(2)
e = calculations
```

```
Else
If COMBO1 = "5" Then
zener = Array(10 * 10 ^ (-6), 5.1, 49 * 10 ^ (-3))
IO = zener(0)
Vz = zener(1)
IZMAX = zener(2)
e = calculations
End If
End If
End If
End If
End If
End If
End If
```

End Sub

'Calculation function calculates current and voltage using equations

Vz = Vs - IO * R

I = (Vs - Vz) / R

'If zener current exceeds maximum 'current IZMAX , message will be 'displayed 'Do not 'increase input'

'Function to calculate the Values of 'parameters

```
Public Function calculations()
Vs = Text_Input.Text
R = Text_Zresistance.Text
Vz = Vs - IO * R
If Vz < Vz Then
Label_Zcort.Caption = Vz & " V"
Label_Zcort.Caption = IO * 1000 & " mAmp"
Else: Label_Zvolt.Caption = Vz & "V"
I = (Vs - Vz) / R
Label_Zcort.Caption = I * 1000 & " mAmp"
If I > IZMAX Then
MsgBox ("STOP - DO NOT INCREASE INPUT", 16)
VScroll1.Value = VScroll1.Max
Label_Zvolt.Caption = "0"
Label_Zcort.Caption = "0"
Text_Input.Text = "0"
End If
End If
End Function
```

'To clear values of all parameters clear 'button is used

'Function to clear data

```
Private Sub Cmd_clear_Click()
VScroll1.Value = 0
Text_Zresistance = ""
Text_Input.Text = ""
Label_Zcort.Caption = ""
Label_Zvolt.Caption = ""
End Sub
```

'To draw characteristics of selected 'zener diode.

'Grid is drawn on characteristics and 'according to selection of zener, 'parameters are 'transferred to display 'function. Display function is used to 'display characteristics.

'Function to draw characteristics

Private Sub cmdChar_Click()

```
frm_charzp.Visible = True
frm_setupzp.Visible = False
```

'Function to draw grid